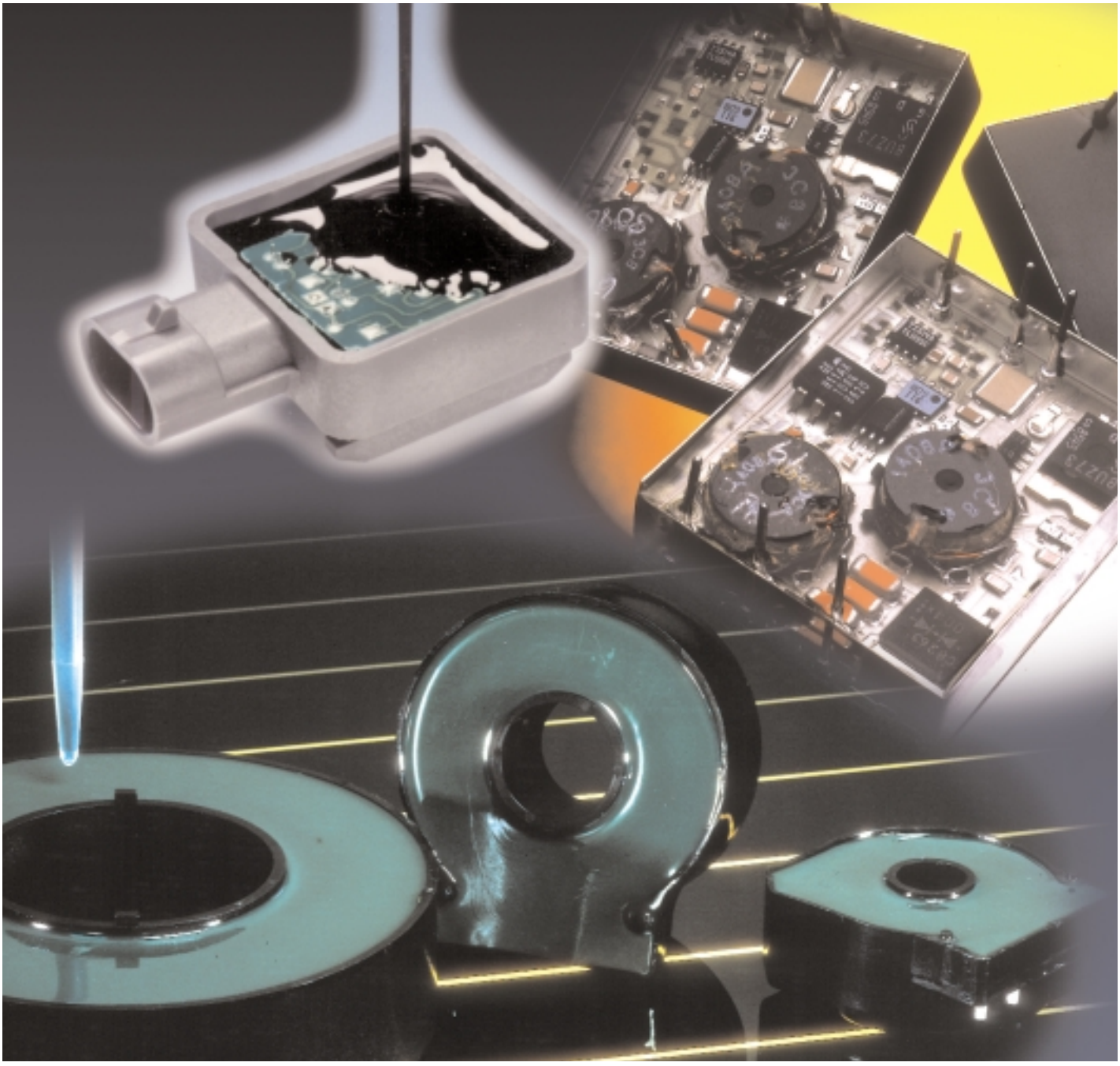
STYCAST® Encapsulants

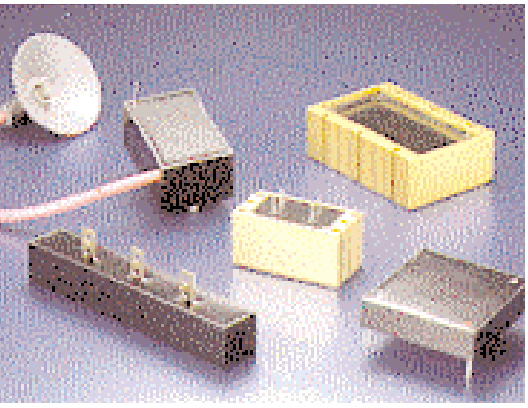
meeting today's performance challenges



Emerson & Cuming
a National Starch & Chemical Company

Encapsulants

STYCAST one component encapsulants provide mechanical and electrical protection for coil windings of all sizes.



In addition to electrical insulation and thermal performance, STYCAST encapsulants protect and support lead/wire connections.

Meeting Today's Performance Challenges

Developed to meet exacting performance demands, STYCAST® encapsulants are designed to protect your assemblies from the degrading environmental and operation effects of chemical exposure, moisture, thermal cycle/shock, current leakage, and mechanical shock and vibration. The embedment of a device can be accomplished by a number of different methods including encapsulation, potting, coating, sealing, impregnation, conformal coating and transfer molding. The embedment of electronic components with STYCAST encapsulants can eliminate the effects of corrosion and minimize the problem of environmental circuit degradation.

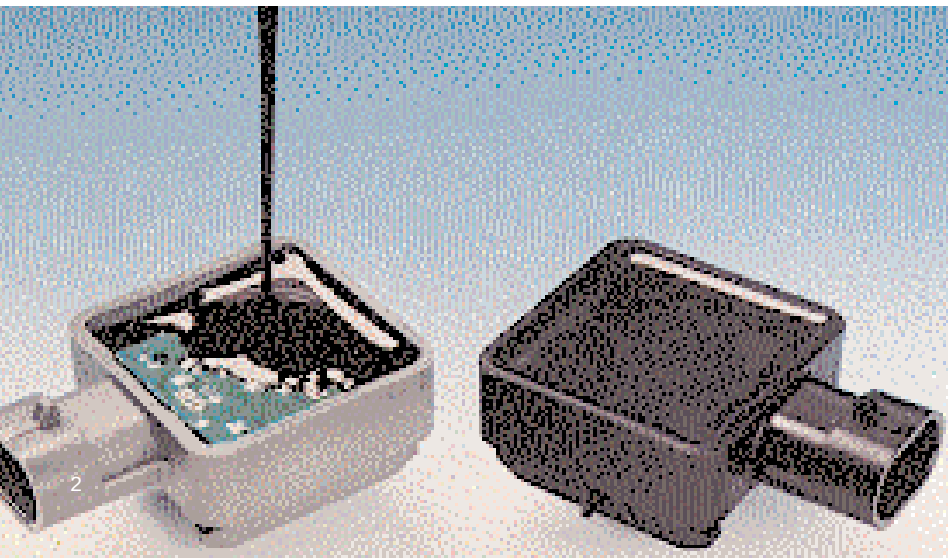
Emerson & Cuming's one- and two-component systems are available from inventory with a range of cure options which can be tailored to meet specific electrical, mechanical and thermal requirements. Certified products are designed to meet military specifications; others have UL® approval for fire resistance; others meet Medical/USP Class VI requirements.

PRODUCTS THAT ARE HELPING TO SOLVE TODAY'S PROBLEMS

STYCAST® one- and two-component encapsulants are specially designed for electrical/electronic applications. Formulated from epoxy, polyurethane, acrylate (UV curable) and silicone chemistries, these systems have proven reliable for electronic insulation because they offer:

- *Excellent elevated temperature stability and thermal shock resistance.*
- *Minimal shrinkage, and stress to critical components during cure.*
- *Outstanding electrical insulation at both room and elevated temperature.*
- *A wide range of handling and physical properties including working life, cure schedule, viscosity, thermal conductivity and hardness.*

Consult Technical Service for detailed information on handling properties and applications for these systems, or contact your local sales engineer for assistance.



STYCAST® encapsulants are designed to protect sensitive electronics from automotive fluids at under-the-hood temperatures.

TYPICAL STYCAST® ENCAPSULANT APPLICATIONS



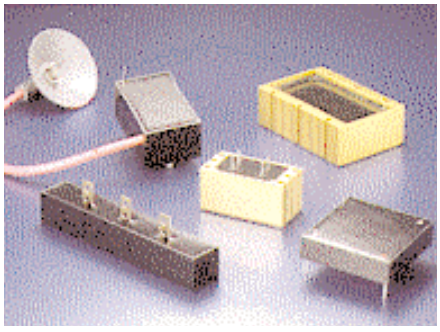
Characteristics to Consider When Selecting Encapsulants

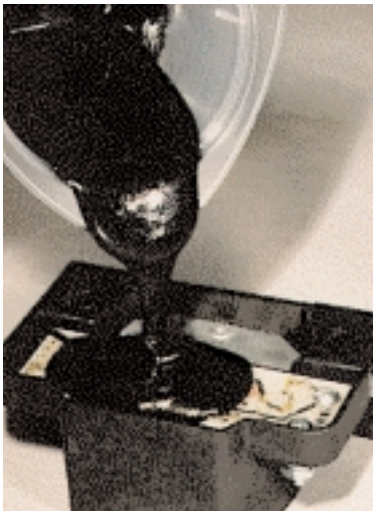
Typical Characteristics of Encapsulant Chemistries

Property	Acrylate	Epoxy	Polyurethane	Silicone
One-Component	✓	✓		✓
Two-Component	✓	✓	✓	✓
Processability	✓	✓	✓	✓
Fast Processing Time	✓			
Repairability	✓		✓	✓
High Temperature Performance		✓		✓
Low Temperature Performance			✓	✓
Flexibility	✓	*	✓	✓
Adhesion	✓	✓	✓	
Thermal Cycle/Shock Resistance	✓	✓	✓	✓
Water Resistance	✓	✓		✓
Chemical Resistance		✓	✓	
Abrasion Resistance	✓		✓	
Low Cost		✓	✓	

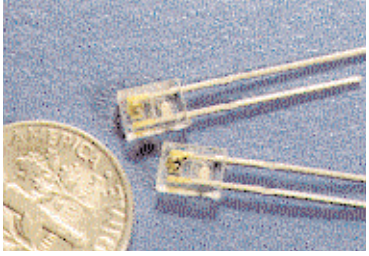
* *The Emerson & Cuming Research team has developed a series of flexible, epoxy based material for stress sensitive electronic applications where the only solutions have historically been silicone. For additional information, please contact Emerson & Cuming Technical Services.*

STYCAST® encapsulants provide mechanical and acoustical properties for a wide range of sensor applications.





STYCAST encapsulants offer a new alternative for protection of components ranging from ASICS to capacitors in applications previously requiring metal cases.



These Light Emitting Diodes are encapsulated with STYCAST® materials formulated for specific optical performance characteristics.

Typical Characteristics of Encapsulant Chemistries

Chemistry	Advantages	Disadvantages	Description
Epoxy	• Range of cure temperatures from below room temperature to 400°F • Low shrinkage • Wide range of viscosities • Excellent adhesion • Minimal or no volatiles during cure • Thermal stability to 450°F • Excellent resistance to environments and chemicals • Excellent electrical properties	• Brittleness at low temperature • Difficult to repair	Offer the widest range of handling and cured properties.
Silicones	• Operational from – 150° to 600°F • Little or no exothermic heat rise during cure • Excellent flexibility • Good electrical properties • Outstanding weather resistance • Low toxicity • High purity	• Low strength • May require primer for good adhesion • Attack by solvents	Unlike epoxies, are elastomeric materials.
Polyurethanes	• Excellent abrasion resistance • Excellent adhesion • Good electrical properties • High impact strength • Excellent low temperature flexibility	• Limited service temperature of 250°F • Poor weatherability • Attack by some solvents	Selected where the flexibility of silicone is required, but cost is a factor.
Acrylates or UV Curable Systems	• One-component • Little or no heat required • 100% solids/no solvents • Cures within 1-20 seconds • Energy savings	• Thin section cure limitations (less than 0.50") • Poor cure in shadowed areas	Offer rapid cure cycles as they cure within seconds when exposed to UV light.

STYCAST encapsulants provide high temperature resistance and low coefficient of thermal expansion for a variety of applications.



The following page presents a sampling of technology solutions. For additional information, or alternative solutions, please contact your nearest Emerson & Cuming representative.

STYCAST® TECHNOLOGY SOLUTION GUIDE

STYCAST Encapsulants	One Part	Two Part	Flame Retardant	Viscosity 25°C cP	Specific Gravity	Temperature Range of Use °C	Hardness Shore D (unless otherwise specified)
Electronic							
XT 5038-6/B 100		✓	✓	3,400	1.49	-40 to +130	85
2651-40/Cat 11		✓	Available	4,000	1.45	-55 to +155	88
933-48	✓			125,000	1.55	-40 to +180	86
1267		✓		450	1.15	-65 to +105	80
E 1000		✓		1,500	1.05	-40 to +65	75 (Shore 00)
Electrical							
906-1	✓			190,000	2.30	-40 to +180	90
TU 906	✓			3,100	1.0	-40 to +130	50 (Shore A)
2850 FT/Cat 11		✓	Available	64,000	2.29	-55 to +155	92
E 151-8	✓			1,100	1.05	-40 to +130	55
1494		✓		8,000	1.82	-40 to +155	90
Sealants							
926-82-1	✓			130,000	1.53	-40 to +180	85
T 640		✓		3,000	1.09	-40 to +105	75
G 508-1	✓		✓	34,000	1.56	-20 to +180	89
A 316	✓			38,000	1.38	-40 to +155	88
Mechanical							
2651 MM/Cat 9		✓		14,000	1.58	-40 to +130	88
4952/Cat 50		✓		35,000	2.2	-65 to +260	70 (Shore A)
1090/Cat 9		✓		30,000	0.79	-40 to +130	80
2057/Cat 23 LV		✓	Available	2,000	1.49	-65 to +105	85

CONVERSIONS

Viscosity

centipoise	Poise	Pa.s = N.s/m²	lb./ft.s
1	0.01	0.001	6.721×10^{-4}
100	1	0.1	0.06721
1000	10	1	0.6721
1487.9	14.88	1.488	1

Thermal Conductivity W/m°K	Dielectric Constant @ 1 mHz	Dielectric Strength V/mil	Comments
0.4	4.2	400	Low viscosity. Meets UL-94-VO.
0.6	3.8	450	Available certified to MIL-I-16293.
0.6	4.0	400	Low CTE Class H potting compound.
NA	3.0	400	Optically clear LED encapsulant.
NA	4.9 (100 Hz)	300	Flexible material, designed for high shock thermal stress applications.
0.86	NA	400	Designed for small devices, excellent thermal shock and heat dissipation.
NA	NA	NA	Tough, flexible abrasion resistant urethane.
1.28	5.4	380	Low stress, high heat dissipation.
NA	NA	325	Semi-flexible, excellent thermal shock and impact resistant.
NA	5.35	450	Best combination of electrical and mechanical properties.
0.6	NA	350	Superior thermal shock and chemical resistance.
NA	NA	NA	Medical grade passes USP Class VI.
0.6	NA	NA	Low CTE, excellent resistance to water, freon and other solvents.
0.5	NA	NA	Exceptional thermal stability and chemical resistance.
0.6	4.4	450	Low viscosity, low abrasion, superior machinability.
1.0	5.2	550	Thermally conductive RTV silicone.
0.19	2.7	375	Low density syntactic foam.
0.4	4.5	400	Designed for easy and rapid air release in casting applications.

Volume Equivalents

in^3	ft^3	U.S. gal	liters	m^3
1	5.787×10^{-4}	4.329×10^{-3}	1.639×10^{-2}	1.639×10^{-5}
1.728×10^{3}	1	7.481	28.32	2.832×10^{-2}
2.31×10^{2}	0.1337	1	3.785	3.785×10^{-3}
61.03	3.531×10^{-2}	0.2642	1	1.000×10^{-3}
6.102×10^{4}	35.31	264.2	1000	1

CONVERSIONS *(continued)*

Pressure

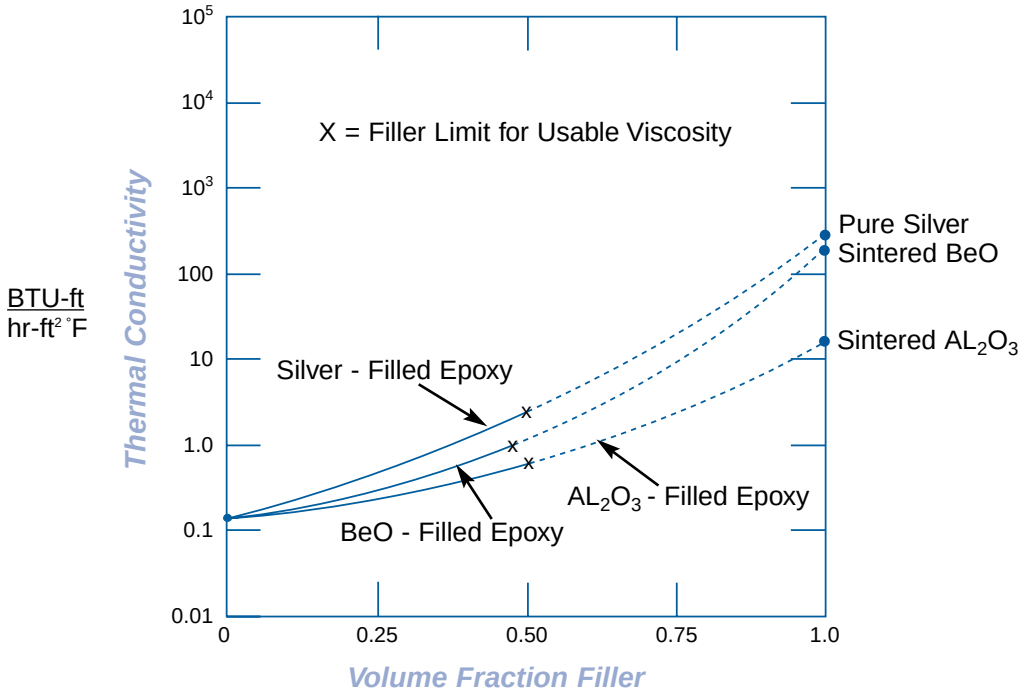
mm Hg	in Hg	bar	atm	N/m²
1	3.937×10^{-2}	1.333×10^{-3}	1.316×10^{-3}	1.333×10^{-2}
25.40	1	3.387×10^{1}	3.342×10^{-2}	3.387×10^{3}
750.06	29.53	1	0.9869	1.000×10^{5}
760.0	29.92	1.013	1	1.013×10^{5}
7.502×10^{-2}	2.954×10^{-4}	1.000×10^{-5}	9.872×10^{-6}	1

Thermal Conductivity

W/m °C	BTU-in / hr-ft² - °F	BTU-ft / hr-ft² - °F	Cal-cm / sec-cm²-°C
1	6.93	0.578	0.002357
0.1442	1	0.0833	0.000344
1.73	12	1	0.004134
418.7	2902.9	241.9	1

THERMAL CONDUCTIVITY VERSUS FILLER

Thermal conductivity measurements vary from one method to another. This graph provides a convenient method for estimating thermal conductivity on a comparative basis. Emerson & Cuming does not use BeO in any products. It is noted in the graph simply as a point of reference.



AMICON® Glob Tops

AMICON® Surface Mount Adhesives

ECCOBOND® Adhesives

ECCOCOAT® Coatings

STYCAST® Encapsulants



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